

Artificial Intelligence CSCC-302

Week-1

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What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a multidisciplinary field that involves the development of computer systems and algorithms with the capacity to perform tasks and make decisions that typically require human intelligence.

Definitions of AI can be categorized into two main dimensions:

Thought processes and reasoning vs. behavior

Fidelity to human performance vs. rationality

Thought Processes and Reasoning vs. Behavior

Thought Processes and Reasoning: This dimension of AI focuses on creating computer systems that can mimic or emulate human-like thinking processes, problem-solving, and decision-making. These AI systems aim to replicate cognitive functions such as logical reasoning, planning, and learning.

Behavior: In contrast, AI can also involve the development of systems that exhibit intelligent behavior without necessarily replicating human thought processes. These AI systems aim to perform tasks, interact with their environment, and produce outcomes that appear intelligent, even if they do not emulate human reasoning.

Fidelity to Human Performance vs. Rationality

AI systems can be evaluated based on two criteria:

Fidelity to Human Performance: This criterion assesses how closely an AI system can replicate the performance of humans in a given task or domain. It measures how well the AI system matches human capabilities, often serving as a benchmark for evaluating AI's success.

Rationality: Rational AI systems aim to make decisions and take actions that optimize outcomes based on defined criteria. Rationality focuses on achieving the best possible result or the best expected result, even if it diverges from human-like behavior.

AI encompasses a broad spectrum of approaches and goals, ranging from replicating human thought processes to exhibiting intelligent behavior. These approaches can be further categorized based on their fidelity to human performance and their emphasis on rational decision-making. AI's versatility allows it to address various challenges and applications across different domains.

Four Approaches to AI

1. Acting Humanly: The Turing Test Approach

- Proposed by Alan Turing in 1950.
- Seeks to define intelligence operationally by indistinguishability from humans.
- The Turing Test involves a computer trying to pass as human through natural language processing, knowledge representation, automated reasoning, machine learning, computer vision, and robotics.
- AI researchers focus less on passing the Turing Test today, emphasizing the study of underlying intelligence principles.

2. Thinking Humanly: The Cognitive Modeling Approach

- Requires understanding human thought processes.
- Achieved through introspection (self-reflection) or psychological experiments.
- Cognitive science combines AI models and psychology to develop precise theories of human cognition.
- AI and cognitive science inform each other, particularly in vision and natural language processing.

3. Thinking Rationally: The "Laws of Thought" Approach

- Inspired by Aristotle's attempt to codify "right thinking."
- Involves precise notations for logical reasoning.
- Logician tradition aims to build intelligent systems based on formal logic.
- Challenges include formalizing uncertain knowledge and dealing with computational complexity.

4. Acting Rationally: The Rational Agent Approach

- Focuses on rational decision-making and behavior.
- Rational agents act to achieve the best outcome or the best expected outcome.
- Rationality involves more than correct inference; it includes adapting to changing environments.
- Rational agents require knowledge representation, reasoning, learning, and perception.

Systems that think like humans	Systems that think rationally
“The exciting new effort to make computers think ... <i>machines with minds</i>, in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ...” (Bellman, 1978)	“The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Systems that act like humans	Systems that act rationally
“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	“Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i>, 1998) “AI ... is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)
Some definitions of artificial intelligence, organized into four categories.	

Advantages of the Rational Agent Approach:

- More general than the "laws of thought" approach.
- Amenable to scientific development.
- Rationality is clearly defined and general, while human behavior is context-specific.

Limitations of Perfect Rationality:

- Achieving perfect rationality in complex environments is often infeasible due to computational demands.
- The working hypothesis in AI is to start with perfect rationality, simplifying the problem and providing foundational principles.
- Later chapters address limited rationality—acting appropriately when computational resources are limited.

AI encompasses various approaches, from mimicking human behavior to formalizing rational thinking. The rational agent approach is a central focus, emphasizing the pursuit of optimal outcomes while acknowledging the challenges of achieving perfection in real-world environments.

Foundation of AI

These contributions from various fields collectively form the foundation of artificial intelligence, providing the theoretical and practical tools needed to address key questions and challenges in AI research.

Philosophy:

- Logic and Methods of Reasoning: Aristotle's laws of rational thinking.
- Mind as a Physical System: Descartes' dualism concept, distinguishing mind and matter.
- Foundations of Learning: Empiricism and induction in understanding knowledge.
- Language and Rationality: Link between language and thought processes.

Mathematics:

- Formal Representation and Proof: George Boole and Gottlob Frege's contributions to formal logic.
- Algorithms and Computability: Alan Turing's concept of computability and Church-Turing thesis.
- (Un)Decidability: Kurt Gödel's work on undecidability in mathematical logic.
- (In)Tractability: Cook and Karp's NP-completeness theory for intractable problems.
- Probability: Foundations of handling uncertainty in AI.

Economics:

- Utility: Utility theory for rational decision-making.
- Decision Theory: Framework for making choices to maximize payoffs.
- Game Theory: Addressing competitive decision-making scenarios.
- Operations Research: Handling sequential decision problems.

Neuroscience:

- Physical Substrate for Mental Activity: Understanding how the brain processes information.
- Localization of Brain Functions: Paul Broca and Santiago Ramon y Cajal's work on brain areas.
- Brain Activity Measurement: EEG, fMRI, and neuron activity recording.
- Challenges in Understanding the Brain: The complexity of brain processes.

Psychology:

- Phenomena of Perception and Motor Control: Hermann von Helmholtz and Wilhelm Wundt's contributions.
- Experimental Techniques: Advancements in controlled experiments.
- Behaviorism vs. Cognition: Transition from behaviorism to cognitive psychology.
- Cognitive Science: Integration of computer modeling into psychology.

Computer Engineering:

- Building Fast Computers: Evolution of computing devices.
- Hardware and Software: Advancements in computer technology.
- Foundations of AI Systems: Hardware and software platforms for AI development.

Control Theory:

- Designing Systems: Designing systems for optimizing objective functions over time.
- Cybernetics: Norbert Wiener's influence on AI and control theory.

Linguistics:

- Knowledge Representation: Relationship between language and thought.
- Grammar and Language Processing: Noam Chomsky's generative grammar theory.
- Computational Linguistics: Application of linguistic principles to AI.

History of AI: Milestones and Developments

This historical timeline highlights key moments in the evolution of AI, from its early conceptualization to recent advancements, demonstrating its increasing impact on various aspects of human life and technology

1943: McCulloch and Pitts introduce the Boolean circuit model of the brain, laying the foundation for neural network concepts.

1950: Alan Turing's groundbreaking paper, "Computing Machinery and Intelligence," introduces the Turing Test and explores machine intelligence.

1952-1969: A period marked by early AI experimentation, exploring various AI concepts and applications.

1950s: Early AI programs emerge, including Samuel's checkers program, Gelernter's Geometry Engine, and Newell & Simon's Logic Theorist and General Problem Solver.

1956: Dartmouth Meeting marks the official adoption of the term "Artificial Intelligence" (AI) and the birth of AI as a field of study.

1965: Robinson develops a complete algorithm for logical reasoning, advancing AI's logical foundations.

1966-1974: AI begins to understand computational complexity, and neural network research experiences a decline.

1969-1979: Early development of knowledge-based systems, emphasizing the role of knowledge representation and reasoning.

1971: Terry Winograd's Shrdlu dialogue system showcases early natural language understanding capabilities.

1980-1988: The expert systems industry sees rapid growth, with AI applications expanding into various domains.

1988-1993: The expert systems industry faces a downturn, often referred to as the "AI Winter," due to unmet expectations and overhyped promises.

1985-1995: Neural networks make a resurgence in popularity, leading to advancements in machine learning.

1988-Now: A resurgence in probability theory, leading to greater technical depth, along with developments in Artificial Life (ALife), Genetic Algorithms (GAs), and soft computing.

1995-Now: The proliferation of intelligent agents in various applications, marking an era of pervasive AI.

1997: IBM's Deep Blue computer defeats the World Chess Champion, Garry Kasparov, demonstrating the potential of AI in strategic games.

2001-Now: The era of very large datasets, with the Google Gigaword Corpus and Wikipedia as notable examples.

2003-Now: Human-level AI becomes a prominent research agenda, aiming to achieve human-like intelligence in machines.

2011: IBM's Watson wins Jeopardy, showcasing AI's ability to process and understand natural language and large volumes of information.

2012: The U.S. state of Nevada permits the testing of driverless cars, a significant milestone in AI and autonomous systems.

Strong AI and Weak AI

"Strong AI" and "Weak AI" are terms used to classify the capabilities of artificial intelligence systems based on their level of consciousness and human-like understanding. Here's a breakdown of these concepts:

1. Strong AI (Artificial General Intelligence - AGI):

- Strong AI refers to artificial intelligence that possesses general intelligence similar to human intelligence.
- It can understand, learn, and apply knowledge across a wide range of tasks and domains, just like a human.
- Strong AI has consciousness, self-awareness, and the ability to think and reason like a human being.
- It has the potential to perform any intellectual task that a human can, and it can adapt to new and unfamiliar situations.
- Achieving strong AI is often considered the ultimate goal of AI research, but it remains a theoretical concept and has not been realized yet.

2. Weak AI (Artificial Narrow Intelligence - ANI):

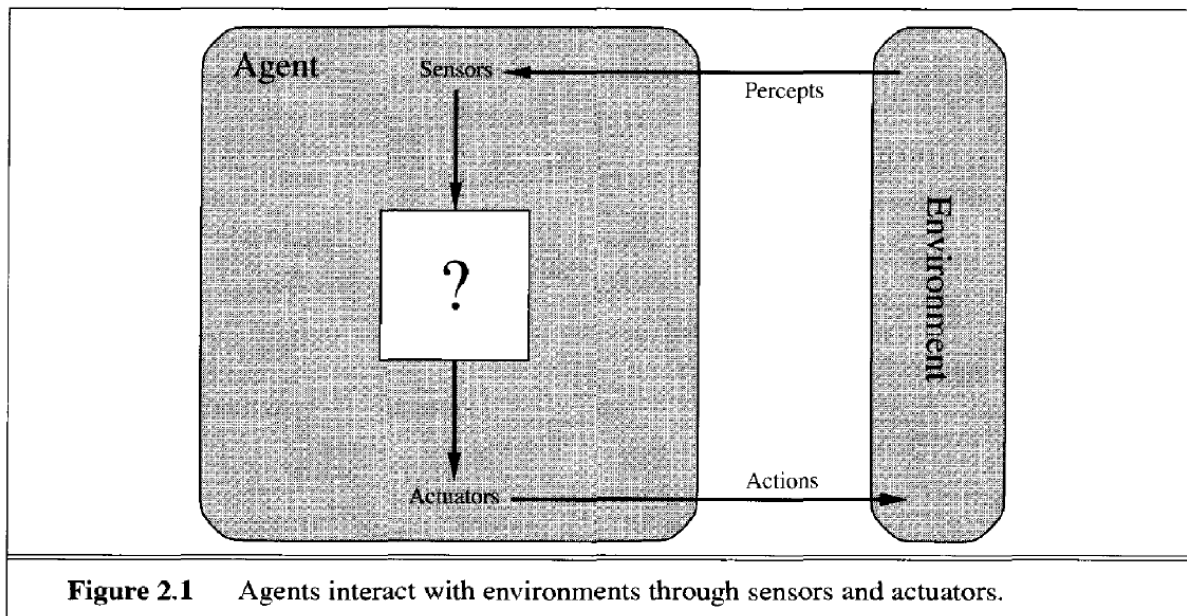
- Weak AI, also known as Narrow AI, refers to artificial intelligence that is specialized in performing specific tasks or solving particular problems.
- It does not possess general intelligence or consciousness. Instead, it operates within a limited scope.
- Weak AI systems are designed to excel in specific domains, such as image recognition, natural language processing, or playing board games like chess or Go.
- These AI systems simulate human-like intelligence for specific tasks but do not have true understanding or awareness.
- Most of the AI applications we encounter today, like virtual assistants and recommendation systems, fall under the category of weak AI.

The primary distinction between strong AI and weak AI lies in their capabilities. Strong AI aims to replicate human-level intelligence, consciousness, and general problem-solving abilities, while weak AI is designed for specialized tasks and lacks human-like understanding. Currently, weak AI is the dominant form of AI in use, while strong AI remains a theoretical concept and a long-term goal in the field of artificial intelligence research.

- The term "percept" refers to the agent's sensory inputs at a specific moment.
- An agent's "percept sequence" is the complete history of all percepts it has ever received.
- An agent's choice of action at any given time can depend on its entire percept sequence.
- The agent's behavior is described by an "agent function" that maps percept sequences to actions.

Agent Function and Agent Program

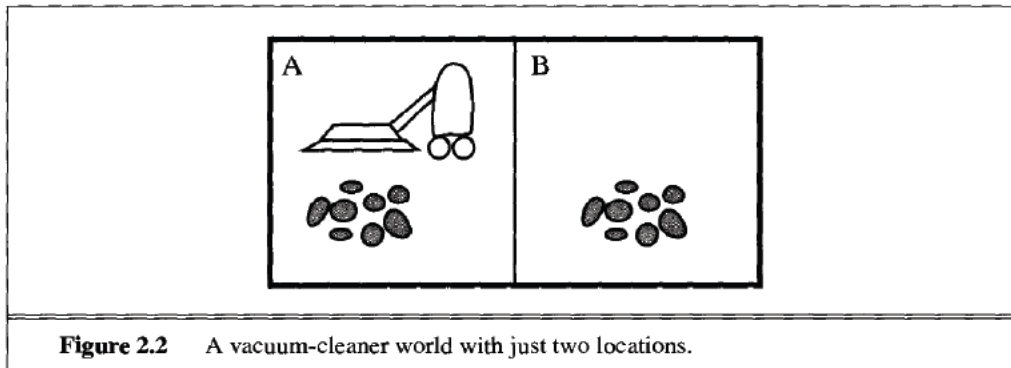
- The agent function is a mathematical description of the agent's behavior.
- It specifies the action an agent takes for any possible percept sequence.
- Tabulating the agent function would require an extremely large table, potentially infinite.
- The agent program, on the other hand, is a concrete implementation that runs on the agent's architecture.
- The program is the internal realization of the agent function.



Illustrative Example: The Vacuum-Cleaner World

- To illustrate these concepts, consider the "vacuum-cleaner world."
- This simple world has two locations: squares A and B, which can be dirty or clean.

- The vacuum agent perceives its location and dirt status and can move left, right, suck dirt, or do nothing.
- A simple agent function for this world: If the current square is dirty, then suck, otherwise move to the other square.
- A partial tabulation of this agent function can be created.



Percept sequence	Action
[A, Clean]	Right
[A, Dirty]	Suck
[B, Clean]	Left
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Right
[A, Clean], [A, Dirty]	Suck
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	Right
[A, Clean], [A, Clean], [A, Dirty]	Suck
⋮	⋮

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Designing an Agent

- The crucial question: What defines a good or bad agent? What makes an agent intelligent or stupid?
- The answer to this question lies in how an agent's function is defined.
- Different ways of filling out the agent function table lead to different agent behaviors.
- Agent design involves making decisions about how to map percepts to actions effectively.

Rationality

- Rationality is a crucial concept when designing agents in artificial intelligence.
- A rational agent is one that consistently makes the right decisions based on the information it has.
- Rationality is tied to the idea of performing actions that lead to the agent's success, but what defines success and rationality?

Performance Measures

- A performance measure defines the criterion for success in an agent's behavior.
- Agents generate sequences of actions based on their percepts from the environment.
- A desirable sequence of actions leads to success, as defined by the performance measure.
- Objective performance measures are preferred, designed by the agent's creator, rather than relying on subjective judgments.

Defining Success in the Vacuum-Cleaner World

- In the vacuum-cleaner world, performance can be measured by factors like the cleanliness of the floor, electricity consumption, and noise generation.
- It's essential to align the performance measure with the actual desired outcomes in the environment.
- The selection of a performance measure is not always straightforward and can have philosophical implications.

Rationality Defined

Rationality depends on four key factors:

- The performance measure for success.
- The agent's prior knowledge of the environment.
- The available actions the agent can perform.
- The agent's percept sequence observed to date.

A rational agent chooses actions that are expected to maximize its performance measure based on the evidence provided by percepts and built-in knowledge.

Rationality vs. Omniscience

- Rationality doesn't require omniscience, as agents can't know the actual outcomes of their actions.
- Percepts and prior knowledge guide rational choices.
- The example of crossing a street illustrates rationality—making the best decision based on available information.

Information Gathering and Learning

- Rational agents may perform actions to gather information or learn from their experiences.
- Agents should adapt their behavior based on new knowledge.
- Agents with complete prior knowledge may not need to perceive or learn but are less flexible.

Autonomy

- Autonomy is the degree to which an agent relies on its own percepts and learning rather than prior knowledge from its designer.
- Rational agents should strive for autonomy.
- Autonomy becomes more critical as agents gain more experience.

Conclusion

- Rational agents make decisions based on their objectives and knowledge.
- Performance measures define what success means for an agent.
- Rationality is distinct from omniscience, emphasizing decisions based on available information.
- Information gathering and learning are essential components of rationality.
- Autonomy is the degree to which an agent relies on its own learning and percepts, promoting adaptability.

The nature of the environment

Introduction

- Rational agents are designed to solve problems or tasks.
- Task environments define the problems or challenges that rational agents are designed to address.
- This lecture focuses on how to specify task environments, covering various properties and dimensions that affect agent design.

Specifying the Task Environment

- Task environments consist of several key components: Performance measure, Environment, Actuators, and Sensors (PEAS).
- Performance measure: Defines what constitutes success for the agent. It could involve multiple, sometimes conflicting, goals.
- Environment: Describes the surroundings in which the agent operates, including roads, other vehicles, pedestrians, and more.
- Actuators: The mechanisms or tools the agent uses to interact with the environment, such as accelerator, brakes, and communication devices.
- Sensors: The sensors provide the agent with information about the environment, including location, speed, and other relevant data.

Example: Automated Taxi Driver

- An example of a complex task environment is an automated taxi driver.
- The PEAS description for the taxi's task environment includes multiple performance goals and considerations.
- The environment involves various types of roads, traffic, weather, and passenger interactions.
- Actuators and sensors are similar to those available to human drivers, with additional capabilities like GPS and distance sensors.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

Other Examples

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, minimize costs, lawsuits	Patient, hospital, staff	Display questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display categorization of scene	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Maximize purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Maximize student's score on test	Set of students, testing agency	Display exercises, suggestions, corrections	Keyboard entry
Figure 2.5 Examples of agent types and their PEAS descriptions.				

Properties of Task Environments

Task environments vary along several dimensions, which influence agent design:

1. Fully observable vs. partially observable.
2. Deterministic vs. stochastic.
3. Episodic vs. sequential.
4. Static vs. dynamic.
5. Discrete vs. continuous.
6. Single agent vs. multiagent.

Fully Observable vs. Partially Observable

- Fully observable environments provide complete information relevant to the agent's decisions.
- Partially observable environments may lack information due to noisy sensors or missing data, impacting decision-making.

Deterministic vs. Stochastic

- Deterministic environments have outcomes solely determined by current states and actions.
- Stochastic environments introduce uncertainty, making predictions challenging.

Episodic vs. Sequential

- Episodic environments divide experiences into independent episodes where decisions do not affect future episodes.
- Sequential environments involve decisions with long-term consequences.

Static vs. Dynamic

- Static environments do not change while agents decide on actions.
- Dynamic environments continuously evolve, requiring real-time decision-making.

Discrete vs. Continuous

- The discrete/continuous distinction applies to state, time, percepts, and actions.
- Discrete environments have finite states and actions, while continuous ones involve continuous values and time.

Single Agent vs. Multiagent

- Single-agent environments involve one agent making decisions.
- Multiagent environments may include interactions between multiple agents, introducing cooperative or competitive aspects.

Conclusion

- Specifying task environments is crucial in designing rational agents.
- Various properties and dimensions, including observability, determinism, and dynamicity, influence agent design and behavior.
- Understanding these properties helps create agents that can effectively tackle a wide range of real-world challenges.
- Environment generators can be used to evaluate agents across different scenarios within an environment class.

The structure of the agent

Introduction

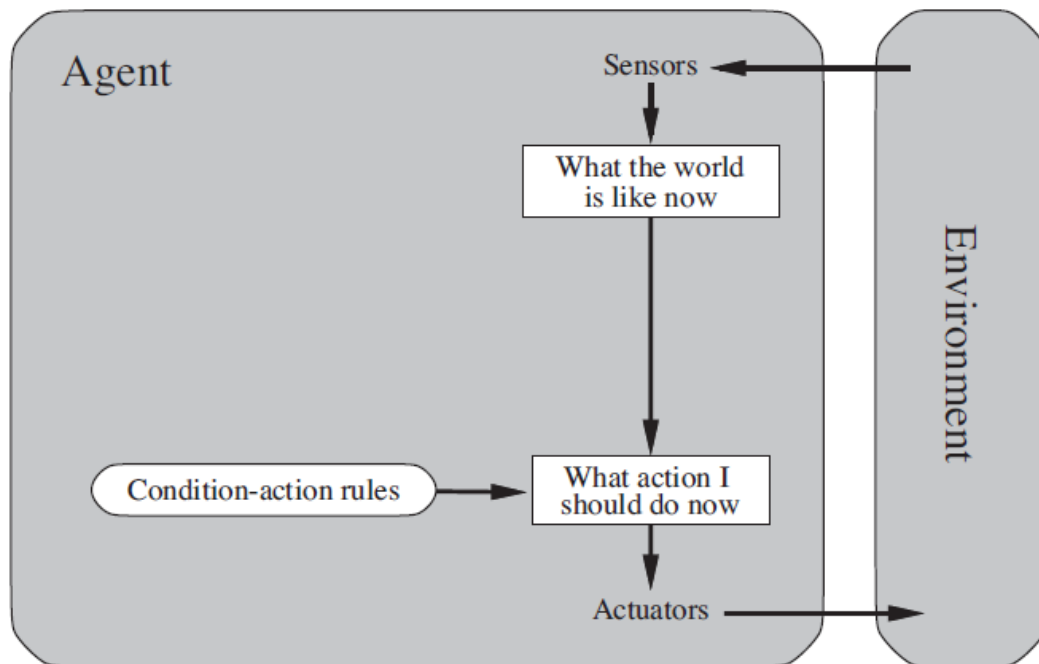
- We've discussed agents based on behavior, but now we'll explore into how they work internally.
- AI's task is to design agent programs that map percepts to actions, which run on computing devices with sensors and actuators (architecture).

Agent Programs

- Agent programs take the current percept as input and return an action to actuators.
- Agent programs differ from agent functions, which consider the entire percept history.
- Using pseudocode language (real programming languages for implementation), we describe agent programs.
- Table-driven agent programs explicitly represent the agent function as a lookup table.
- However, this approach is infeasible due to the vast number of table entries for complex environments.

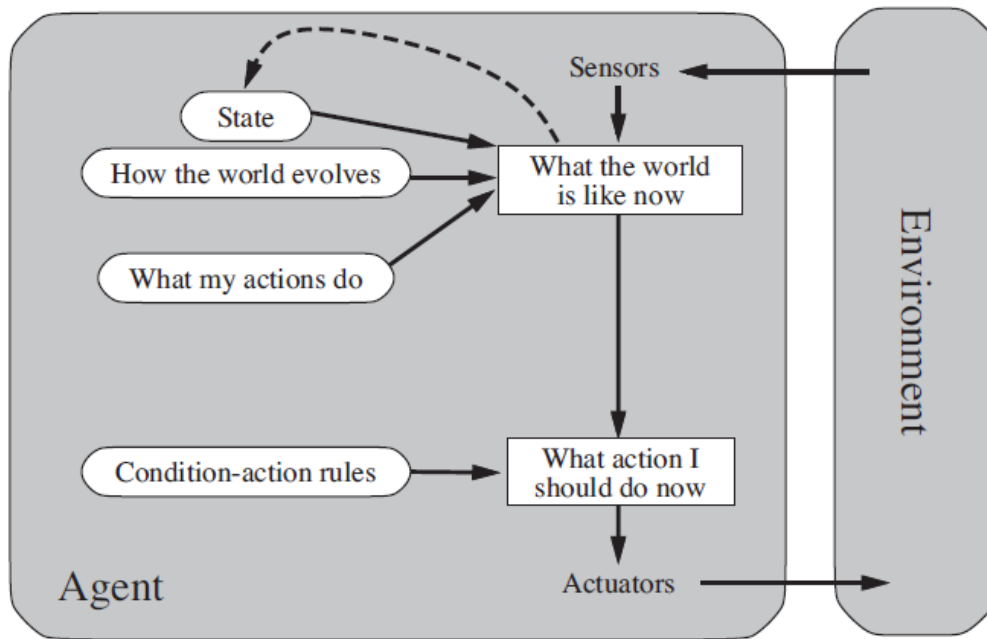
Simple Reflex Agents

- Simple reflex agents base actions solely on the current percept, ignoring percept history.
- They work well in fully observable environments but fail in partially observable ones.
- Condition-action rules, such as "if car-in-front-is-braking then initiate-braking," guide their behavior.
- Randomization can help simple reflex agents escape infinite loops and improve performance.



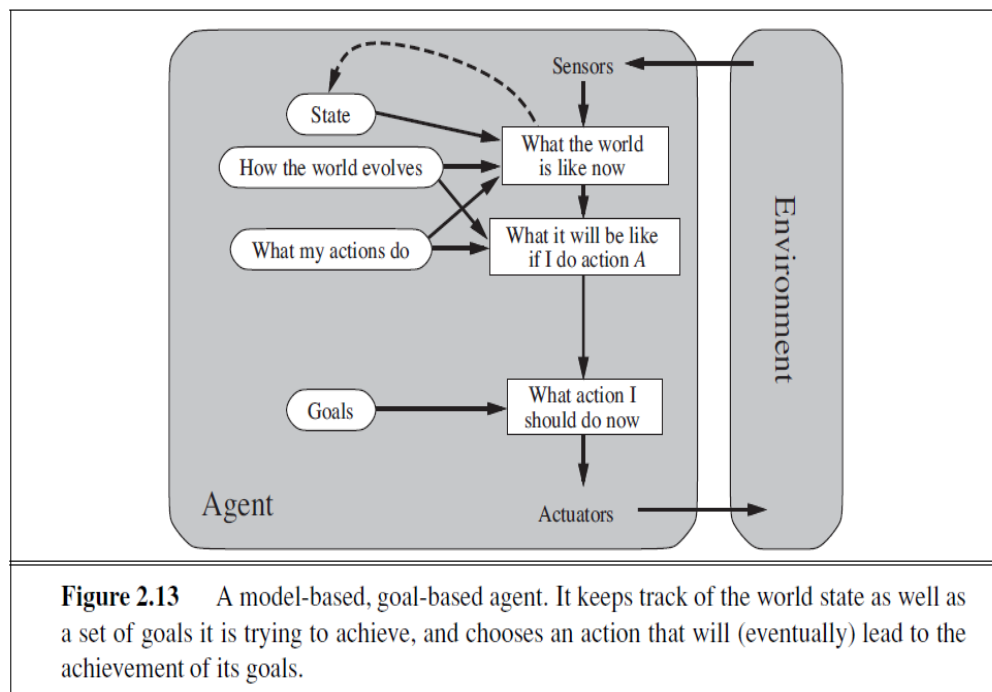
Model-Based Reflex Agents

- Model-based agents maintain internal state to handle partial observability.
- They use knowledge about how the world evolves and how actions affect it to update their internal state.
- These agents are more flexible than simple reflex agents but require a model of the world.



Goal-Based Agents

- Goal-based agents consider not just the current state but also goals when selecting actions.
- They may require planning and search to determine the best course of action.
- This approach allows for more flexible decision-making based on goal satisfaction.



Utility-Based Agents

- Utility-based agents go beyond goals and consider a utility function that assigns values to states or sequences of states.
- Utility functions allow trade-offs between conflicting goals and decision-making under uncertainty.
- Utility-based agents aim to maximize the expected value of their utility function.

Learning Agents

- Learning agents consist of four components: performance element, learning element, critic, and problem generator.
- The learning element modifies the performance element based on feedback from the critic to improve future performance.
- The critic evaluates the agent's performance with respect to a fixed performance standard.
- The problem generator suggests exploratory actions to help the agent learn better behaviors.
- Learning can occur in various forms, including learning from percepts, action outcomes, and utility information.

